Skywalker at BAREC Shared Task 2026: Multi-Expert PLM Fusion and QWK-Aware Ordinal Decoding

Junhong Liang
Mohamed bin Zayed University of Artificial Intelligence
Abu Dhabi, UAE

Xueqi Li
Anton Oilfield Services
Baghdad, Iraq

junhong.liang@mbzuai.ac.ae

Abstract

We present the system of Team Skywalker for the Sentence-level Strict track of the BAREC Shared Task 2026, which requires predicting one of 19 ordered Arabic readability levels for each sentence. Our system integrates complementary public Arabic pretrained language-model (PLM) experts fine-tuned with classification, regression, and ordinal objectives. Their continuous scores, confidence estimates, and rank-based features are combined using cross-fitted LightGBM meta-models. We further incorporate a cumulative ordinal branch that models level-boundary probabilities and a normalized-text branch that improves robustness to Arabic orthographic variation. Finally, a QWK-aware decoder maps the fused continuous score to one of the 19 readability levels using monotonic thresholds selected exclusively on the public validation set. We use only the official training data for supervision and the public validation labels for model selection. In addition to the final fusion system, we report single-PLM baselines and controlled ablations. The final configuration achieves a QWK of 86.30 on the public validation set and 84.20 on the CodaBench blind evaluation, ranking fourth among the submitted systems.

1 Introduction

Readability assessment plays an important role in educational applications, including second-language learning (Xia et al., 2016), translation quality assessment (Imperial et al., 2026), and error-correction feedback (Liang et al., 2026). Arabic readability assessment requires fine-grained judgments of lexical, morphological, syntactic, and discourse-level difficulty. The BAREC Shared Task 2026 requires systems to predict one of 19 ordered readability levels for Arabic text (Elmadani et al., 2026). In this work, we focus on the sentence-level formulation. BAREC covers a broad range of domains, sources, and target audiences (Elmadani et al., 2025b). Its fine-grained annotation scheme makes adjacent readability levels difficult to distinguish, while the annotation guidelines emphasize that readability cannot be determined by a single surface-level property (Habash et al., 2025).

The 19 labels are ordered and fine-grained, which requires reliable discrimination between nearby levels. In addition, variation in Arabic domains and orthographic forms introduces further distributional variation. Because the evaluation metric imposes larger penalties on more distant ordinal errors, we separate representation learning, continuous ranking, and discrete decoding. Our Sentence-level Strict system fuses raw-score, ordinal, and normalized-text predictions and then applies a validation-selected QWK-aware decoder. We report controlled branch ablations and clearly distinguish organizer-provided blind evaluation results from local validation results.

2 Background

The BAREC-ST 2026 shared task evaluates fine-grained Arabic readability prediction (Elmadani et al., 2026). We participate in the Sentence-level Strict track, where the input is an Arabic sentence u and the target is an ordered readability label $y \in \{1, \ldots, 19\}$. In contrast to coarse-grained readability classification, the 19-level formulation requires systems to distinguish between adjacent difficulty levels that are often linguistically similar. Consequently, a prediction error from level 8 to level 9 should be considered substantially less severe than an error from level 8 to level 18.

The official data include sentence text and metadata describing the source, domain, and text class. The dataset contains 54,845 training instances, 7,310 public validation instances, and 7,285 public test instances.

The official evaluation metric is quadratic weighted kappa (QWK), which accounts for the

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ordinal distance between predicted and gold readability levels. The task requires both reliable ranking and appropriate calibration: ***a system must preserve the relative difficulty ordering of sentences while placing decision boundaries appropriately across the 19 discrete levels.*** Our system consequently produces a continuous difficulty score and then applies a validation-selected, monotonic QWK-aware decoder to obtain the final readability level.

3 System Overview

Our system separates readability prediction into two stages. First, it constructs several continuous difficulty estimates from complementary Arabic PLM experts and LightGBM branches. Second, it converts the fused score into a discrete readability level using a validation-selected monotonic decoder. This separation allows the model to optimize sentence ordering and metric-specific ordinal calibration independently.

3.1 Problem Formulation and Arabic Input Views

Each example is denoted by $x = (u, \mathbf{m})$, where u is an Arabic sentence and $\mathbf{m}$ contains the released source, domain, and text-class metadata. Given the official training set $\mathcal{D}_{\text{tr}} = \{(u_i, \mathbf{m}_i, y_i)\}_{i=1}^{N}$, the target label is $y_i \in \mathcal{Y} = \{1, \ldots, K\}$, where $K = 19$. The system learns a continuous score $s(x)$ and maps it to an ordinal label with monotonic thresholds $\mathbf{t}$:

$$\hat{y} = d_{\mathbf{t}}(s(x)), \qquad d_{\mathbf{t}}(z) = 1 + \sum_{k=1}^{K-1} I[z > t_k] \tag{1}$$

where $t_1 < \cdots < t_{K-1}$. Thus, the PLM and LightGBM components estimate relative sentence difficulty, whereas the decoder determines where the 18 boundaries between the required labels should lie.

Raw, normalized, and lookup views. Figure 1 uses three deterministic views of the same sentence. The raw view, $u_{\text{raw}} = u$, is used by the principal PLM path; before tokenization, it receives punctuation cleanup and D3TOK morphological segmentation (Obeid et al., 2020). The normalized view, $u_{\text{norm}} = N(u)$, applies Unicode NFKC normalization, removes tatweel and combining marks, including diacritics, and collapses repeated whitespace. This view is used only by the normalized-text branch.

We also define a stricter lookup key $\kappa(u)$. In addition to the normalization above, $\kappa(\cdot)$ removes Unicode punctuation, symbols, and control characters, maps every Unicode digit to `0`, and collapses whitespace. This key is used exclusively to identify exact matches in the official training data; it is not used to retrieve labels from validation, public-test, or blind-test data.

3.2 PLM Experts and Feature Construction

We use public Arabic PLMs from two complementary expert families. The first family contains four 19-way classification experts: AraBERT v2 (Antoun et al., 2020), MARBERT (Abdul-Mageed et al., 2021), CAMeLBERT-MSA (Inoue et al., 2021), and AraELECTRA (Antoun et al., 2021). The second family contains NeoAraBERT (Abou Chakra et al., 2026) score experts trained with regression, EMD, and multitask objectives.

19-way classification experts. Let $\mathcal{M}_{\text{CE}}$ denote the four classification backbones. For each $m \in \mathcal{M}_{\text{CE}}$, we initialize the encoder from its public pretrained checkpoint. Given a tokenized raw sentence u_{raw}, the encoder produces a pooled representation $\mathbf{h}_m(u_{\text{raw}}) \in R^{d_m}$. We attach a newly initialized linear prediction head:

$$\mathbf{z}_m(u_{\text{raw}}) = \mathbf{W}_m \mathbf{h}_m(u_{\text{raw}}) + \mathbf{b}_m, \tag{2}$$

where $\mathbf{W}_m \in R^{K \times d_m}$ and $\mathbf{z}_m$ contains one logit for each readability level. Both the encoder and prediction head are fine-tuned end-to-end on the official BAREC training split.

To compensate for uneven level frequencies, let n_ℓ denote the number of training examples at level ℓ. We use the inverse-frequency weight $w_\ell = \frac{N}{K n_\ell}$.

The probability assigned by expert m to level ℓ is

$$p_m(\ell \mid u_{\text{raw}}) = \frac{\exp(z_{m,\ell}(u_{\text{raw}}))}{\sum_{j=1}^{K} \exp(z_{m,j}(u_{\text{raw}}))}. \tag{3}$$

The corresponding weighted cross-entropy objective is

$$\mathcal{L}_{\text{CE}}^{(m)} = -\frac{1}{N} \sum_{i=1}^{N} w_{y_i} \log p_m(y_i \mid u_{i,\text{raw}}). \tag{4}$$

Here, p_m denotes the softmax output of a BAREC-fine-tuned classifier; it is not a fusion weight. We select each expert checkpoint by public-validation QWK, without updating model parameters on validation examples.

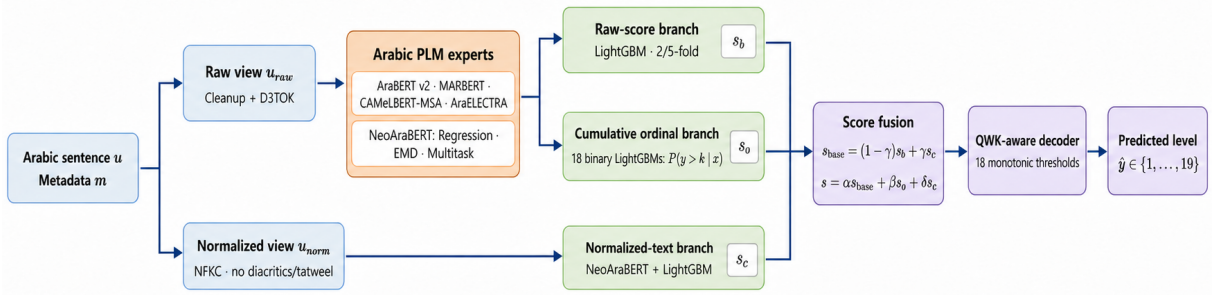


Figure 1: Sentence-level architecture. The raw Arabic sentence is processed by the PLM feature path, whereas a normalized text view is used by a complementary normalized-text branch. A stricter normalized key is used only for a training-only exact lookup. The three branches output continuous difficulty scores, which are fused and decoded into one of 19 ordered levels.

PLM-derived features. For every classification expert, we extract its expected difficulty, normalized entropy, confidence, and split-wise rank:

$$\begin{aligned} \mu_m(u) &= \sum_{\ell=1}^{K} \ell \, p_m(\ell \mid u), \\ \eta_m(u) &= -\frac{1}{\log K} \sum_{\ell=1}^{K} p_m(\ell \mid u) \log p_m(\ell \mid u), \\ c_m(u) &= \max_{\ell} p_m(\ell \mid u). \end{aligned} \tag{5}$$

The rank feature is computed within the corresponding scoring split. We concatenate these summaries across classification experts into $\mathbf{r}_{\text{cls}}$. We do not pass hidden states or complete 19-dimensional probability vectors to LightGBM.

In parallel, NeoAraBERT MSA/mixed-regression, EMD, and multitask experts produce scalar difficulty scores collected in $\mathbf{a}_{\text{score}}$. The multitask expert additionally uses the released 7-, 5-, and 3-level labels during training. The raw feature vector for the LightGBM branches is

$$\begin{aligned} \phi_{\text{raw}}(x) = \big[&\mathbf{r}_{\text{cls}}, \mathbf{a}_{\text{score}}, n_{\text{word}}, n_{\text{char}}, \\ &n_{\text{char}} / \max(1, n_{\text{word}}), \text{onehot}(\mathbf{m})\big]. \end{aligned} \tag{6}$$

3.3 Cross-Fitted Difficulty Branches

All LightGBM branches use cross-fitting on the public validation data. Let $\mathcal{V} = \bigcup_{r=1}^{R} \mathcal{V}_r$ denote a partition of the public validation set. A branch model fitted on $\mathcal{V} \setminus \mathcal{V}_r$ predicts only the held-out fold $\mathcal{V}_r$. Concatenating such held-out predictions yields out-of-fold (OOF) scores for model selection and fusion. At blind inference, the corresponding fold predictors are averaged. This prevents a branch from being evaluated on validation instances used to fit that branch.

Branch I: raw-score / QWK-boundary branch. Two LightGBM regressors are trained from $\phi_{\text{raw}}(x)$ using two-fold and five-fold cross-fitting, producing continuous scores $s_{b,2}(x)$ and $s_{b,5}(x)$. They are blended as

$$s_b(x) = (1-\lambda) s_{b,2}(x) + \lambda s_{b,5}(x). \tag{7}$$

We call this the *QWK-boundary branch* because its standalone version is decoded with QWK-optimized thresholds. Importantly, final fusion uses the continuous score $s_b(x)$ rather than its standalone discrete prediction.

Branch II: cumulative ordinal branch. The ordinal branch explicitly models whether a sentence exceeds each level boundary. For every $k \in \{1, \ldots, K-1\}$, we define the binary target

$$b_{i,k} = I[y_i > k], \tag{8}$$

and fit a LightGBM classifier estimating $q_k(x) \approx P(y > k \mid x)$. The cumulative ordinal score is

$$s_o(x) = 1 + \sum_{k=1}^{K-1} q_k(x). \tag{9}$$

This branch directly encodes the ordered relation between adjacent readability levels, instead of treating all 19 labels as unrelated classes.

Branch III: normalized-text branch. The normalized branch recomputes three NeoAraBERT score sources from u_{norm}. It retains the shallow and metadata features in Equation 6, while replacing the corresponding raw-text NeoAraBERT scores with their normalized-text counterparts. A lower-capacity, two-fold LightGBM regressor then produces $s_c(x)$. This branch provides a complementary estimate that is less sensitive to Arabic diacritics, tatweel, and related surface-form variation.

3.4 Score Fusion, Decoding, and Parameter Selection

We first construct a raw–ordinal base score:

$$s_{\text{base}}(x) = (1-\gamma)s_b(x) + \gamma s_o(x). \quad (10)$$

The final continuous difficulty estimate combines the base score, an additional direct ordinal contribution, and the normalized-text score:

$$s(x) = \alpha s_{\text{base}}(x) + \beta s_o(x) + \delta s_c(x). \quad (11)$$

We perform a multi-stage hyper-parameter selection as described in Appendix D. The selected configuration is $\lambda = 0.90$, $\gamma = 0.275$, and $(\alpha, \beta, \delta) = (0.10, 0.40, 0.50)$. Thus, the effective weights of s_b, s_o, and s_c are 0.0725, 0.4275, and 0.50, respectively. The ordinal score appears twice because it first contributes to the raw–ordinal base score and then receives an independently selected direct contribution in the final fusion.

4 Experimental Setup

Experimental parameters. Detailed Parameter for fine-tuning PLM expert is available in Appendix A and the parameters for training LightGBM regressor is available in Appendix B. All hyperparameters, fusion weights, and decoder thresholds are selected using only the official training data and public validation labels, then frozen before blind inference. We evaluate with QWK, which penalizes more distant ordinal errors more heavily.

Baselines and controlled ablations. Table 1 compares our final system with representative single-PLM baselines of *NeoAraBERT*, *AraBERT v2* and *CAMeLBERT-MSA* with their best public-validation-selected scalar-regression configurations. Table 2 provides controlled ablations of the final system. "w/o normalized-text branch" removes the normalized-text LightGBM score, whereas "w/o ordinal branch" removes the cumulative ordinal score; for both settings, the retained score weights and monotonic thresholds are reselected using public validation labels only. "w/o QWK-aware decoder" retains the full fused score but replaces the validation-optimized monotonic decoder with direct nearest-integer rounding. "w/o score fusion" uses only the raw-score/QWK-boundary expert, while "ordinary raw-score only" uses only the conventional raw LightGBM regression score.

System	Val. QWK $\times 100$	Acc. (%)	Acc.$\pm$1 (%)
NeoAraBERT	82.36	34.9	69.6
AraBERT v2	82.90	34.7	70.2
CAMeLBERT-MSA	82.54	32.7	69.6
Final score fusion	**86.30**	**41.8**	**73.4**

Table 1: Main comparison on the public validation split. Each single-PLM baseline uses a public-validation-selected checkpoint and prediction objective.

Scoring configuration	Val. QWK	Acc. (%)	Acc.$\pm$1 (%)
Final score fusion	**86.30**	**41.8**	73.4
w/o normalized-text branch	86.03	39.8	**73.5**
w/o ordinal branch	86.23	38.1	73.3
w/o QWK-aware decoder	85.04	39.9	73.1
w/o score fusion (raw-score only)	85.98	38.8	72.9
ordinary raw-score only	85.62	38.9	72.6

Table 2: Controlled component ablation analysis on the public validation split.

5 Results

Tables 1 and 2 report complementary validation results. The final system obtains 86.30 QWK, 41.8% exact accuracy, and 73.4% within-one accuracy. Compared with the strongest single-PLM baseline, AraBERT v2 regression, final fusion improves QWK by 3.40 points. The controlled ablations show that all three design choices contribute. Removing the normalized-text branch decreases QWK by 0.27 points, while removing the ordinal branch decreases QWK by 0.07 points and reduces exact accuracy from 41.8% to 38.1%. The largest decrease occurs without the QWK-aware decoder: direct rounding lowers QWK by 1.25 points. Using only the raw-score expert or only the ordinary raw LightGBM score also reduces QWK to 85.98 and 85.62, respectively. These results support score-level fusion and metric-aware ordinal decoding.

The final submitted system obtained 84.20 QWK on the returned CodaBench blind-test evaluation, ranking 4th out of all submissions.

6 Conclusion

We presented Team Skywalker's system for Sentence-level Strict Arabic readability prediction in BAREC 2026. Our approach combines complementary Arabic PLM estimates with raw-score, cumulative ordinal, and normalized-text LightGBM branches, followed by QWK-aware monotonic decoding. The final fusion achieved 86.30 validation QWK and 84.20 QWK on the blind evaluation, showing that combining ordinal modelling, Arabic normalization, and metric-aligned decoding is effective for fine-grained 19-level readability assessment.

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A Classification-Expert Training Configurations

Table 3 reports the shared configuration for the four 19-way classification experts. All values were fixed before final blind inference. We use the official training split for optimization and the public validation split only for checkpoint selection and subsequent fusion calibration.

For a sentence longer than the maximum input length, the tokenizer truncates the sequence to 128 tokens. We use the validation-selected checkpoints to produce the expected-score, entropy, and confidence features consumed by the LightGBM fusion branches.

B LightGBM Training Configurations

Table 4 lists the LightGBM configurations used in our score branches. The raw-score branch is trained under both two- and five-fold cross-fitting, whereas the normalized-text and cumulative ordinal branches use two-fold cross-fitting. Each ordinal model $k \in \{1, \ldots, 18\}$ is a separate binary classifier for $P(y > k \mid x)$.

C Related Work

Arabic readability assessment. Automatic readability assessment estimates whether a text is appropriate for a target reader. Early Arabic approaches adapted formula-based readability measures to Arabic-specific orthographic and syllabic properties (El-Haj and Rayson, 2016). Subsequent work showed the value of richer lexical, morphological, and syntactic features for both Arabic L1 and L2 readers (Saddiki et al., 2018). More recently, DARES introduced Arabic school-material readability prediction at both coarse and fine granularities (El-Haj et al., 2024), while Liberato et al. (Liberato et al., 2024) compared rule-based, feature-based, and Arabic PLM approaches.

BAREC provides a large, balanced resource for fine-grained Arabic readability assessment across 19 levels (Elmadani et al., 2025b). Its annotation guidelines emphasize that readability depends on interacting lexical, morphological, syntactic, pedagogical, and cognitive factors rather than sentence length alone (Habash et al., 2025). The BAREC demo further illustrates the practical use of sentence-level readability prediction for text search and analysis (Altarbouch et al., 2025).

The BAREC Shared Task 2025 is the closest benchmark to our setting (Elmadani et al., 2025a). Participating systems explored model- and objective-diverse transformer ensembles (Basem et al., 2025), ordinal regression with K-fold ensembling (Ayesh, 2025), and uncertainty-aware decoding (Abdou, 2025). Building on this evidence, our system uses cross-fitted score fusion over complementary PLM experts and separates continuous difficulty ranking from the final QWK-aware ordinal decoder.

Arabic PLMs and ordinal prediction. Recent public Arabic PLMs differ in their pretraining data, objectives, and Arabic coverage. Our experts are based on AraBERT v2 (Antoun et al., 2020), MARBERT (Abdul-Mageed et al., 2021), CAMeLBERT-MSA (Inoue et al., 2021), AraELECTRA (Antoun et al., 2021), and NeoAraBERT (Abou Chakra et al., 2026). A standard 19-way classifier can learn useful text representations, but it does not explicitly model the distance between adjacent readability levels. We therefore retain classification experts for complementary semantic and confidence signals, while adding a cumulative ordinal branch and a validation-selected, QWK-aware monotonic decoder for the ordered final decision.

D Validation-only parameter selection

No blind labels are used during system design. We first create a stratified 50/50 partition of the public validation set, preserving the distribution of the 19 labels. For each candidate raw LightGBM ca-

Table 3: Shared hyperparameters for the four 19-way classification experts.

Component	Configuration
Backbones	AraBERT v2, MARBERT, CAMeLBERT-MSA, and AraELECTRA
Prediction space	19 ordered BAREC readability levels
Input view	Raw Arabic sentence
Maximum sequence length	128 tokens
Classification head	Newly initialized linear $d_m \rightarrow 19$ head
Fine-tuned parameters	Full encoder and classification head
Loss	Inverse-frequency weighted cross-entropy
Optimizer	AdamW
Learning rate	2×10^{-5}
Weight decay	0.01
Warmup ratio	0.10
Batch size	16
Maximum epochs	5
Checkpoint criterion	Highest public-validation QWK

Table 4: LightGBM training configurations for the score branches. The ordinal branch contains 18 binary classifiers, one for each cumulative decision $P(y > k \mid x)$.

Hyperparameter	Raw-score branch	Normalized-text branch	Ordinal branch
Objective	Regression L2	Regression L2	Binary classification
Number of leaves	15	15	15
Minimum child samples	150	100	100
Learning rate	0.020	0.025	0.030
Number of trees	350	300	250
L_2 regularization	30	20	20
Column subsampling	0.80	0.80	0.85
Row subsampling	0.85	0.85	0.90
Subsampling frequency	1	1	1
Random seed	123	123	123
Cross-fitting	2- and 5-fold	2-fold	2-fold

pacity, we fit the regressor and its decoder on one half and measure QWK on the other. The selected low-capacity configuration achieves 85.20 held-out QWK, compared with 85.15 and 84.91 for two larger alternatives. The normalized companion, evaluated by the same procedure after replacing three raw NeoAraBERT features with normalized-text features, achieves 85.33.

We then select fusion parameters sequentially. We scan 21 values of λ for the two-/five-fold raw-score blend, retain the four strongest candidates, refit their decoders on the selection half, and select the 0.10/0.90 two-/five-fold mixture. Next, we scan 25 values of γ for the raw–ordinal interpolation in Equation 10. Because this stage selects a continuous ranking before discretization, we use validation Spearman correlation and select $\gamma = 0.275$. Finally, we evaluate 24 feasible (α, β, δ) triples. Each triple receives its own OOF score, training-only lookup, and validation-fitted QWK decoder.

For every decoder fit, the thresholds are initialized in increasing order and optimized coordinate-wise. A threshold update is retained only when it preserves monotonicity and improves validation QWK. After this procedure, the PLM checkpoints, LightGBM branches, fusion weights, lookup rule, and thresholds are frozen before blind inference.